

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E :- Computer Science and Engineering	Semester	: III
Course Name	: Digital Design and Computer Organization	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Using K-Map, simplify the Boolean expression $F(A,B,C,D) = \sum m(0,1,2,3,4,5,12,14) + d(8,10)$ and write the circuit using i) NAND gates only for SOP form ii) NOR gates only for POS form. CO1 (12)
 - Design and implement half and full adder using basic gates. CO1 (08)
- Find the prime implicants, essential prime implicants and minimal Sum of Product for the function $F(A,B,C,D) = \sum m(1,3,5,10,11,12,13,14,15)$ using QM method. CO1 (10)
 - Give the differences between demultiplexer and decoder. Implement the following functions using suitable active high enable demultiplexer and decoder (active high output)
i) $Y1 = F(A,B,C) = \sum m(1,3,4,5,6)$ ii) $Y2 = F(A,B,C) = \sum m(1,2,3,6)$
use appropriate logic gates if necessary. CO1 (10)

UNIT - II

- Construct the SR latch using NAND and NOR gates. Also explain its working. CO2 (10)
 - What is asynchronous counter? Design and implement 3 bit up/down ripple counter using negative edge triggered T flip flops and also write its timing diagram. CO2 (10)
- Write the logic symbol, circuit, truth table and timing diagram of negative edge triggered SR flip flop. CO2 (05)
 - Briefly explain the working of D flip flop. CO2 (10)
 - Design a synchronous counter with counting sequence: 000, 001, 011, 111, 110, 100, 000,...negative edge triggered JK flip flops. CO2 (05)

UNIT - III

- Explain with a neat diagram the basic functional units of a computer. CO3 (05)
 - Analyze Big Endian and Little Endian methods of byte addressing with proper example. CO3 (05)
 - Register R1 and R2 of a computer contain the decimal values 1200 and 6000. Assume that the computer has 32bit word length. Calculate the effective address of memory operand in each of the following instructions. CO3 (10)
 - Load 10 (R1), R5
 - Move #2000, R5
 - Store 30(R1,R2),R5
 - Add (R2)+, R5
 - Subtract - (R1), R5

6. a) Explain how to encode the instructions into 32 bit words with an example. CO3 (08)
 b) What is performance measurement? Explain the overall SPEC rating for the computer in a program suite. CO3 (08)
 c) A program contain 1000 instructions. Out of that 25% instructions require 4 clock cycles, 40% instructions require 5 clock cycles and remaining requires 3 clock cycles for execution. Find the total time required to execute the program running in a 1 GHz machine. CO3 (04)

UNIT- IV

7. a) Write a microroutine for the execution of the instruction: ADD R1,(R2),R3. CO4 (08)
 b) Perform the following using bit-pair multiplication algorithm: i) $-13 * -6$ ii) $18 * -7$ CO4 (06)
 c) With a neat diagram, explain the classic five stage pipeline for RISC Processor. CO4 (06)
8. a) Illustrate the working of Carry Save Addition with 43 and 18 as the Multiplicand and Multiplier. CO4 (08)
 b) Perform $28 / 3$ using Non-restoring division algorithm. CO4 (06)
 c) Identify the hazards in the following statements if any, then provide the solution for the identified hazards. CO4 (06)
 LW R1, 0(R2)
 SUB R4, R1, R2
 ADD R3, R5, R1
 OR R6,R1,R2

UNIT - V

9. a) Define the following with respect to cache memory: i) Hit rate ii) Miss rate iii) Write-through cache iv) Write-back cache. CO5 (06)
 b) Describe the different schemes employed to handle simultaneous interrupt requests. CO5 (07)
 c) Write a short note on enabling and disabling of interrupts. CO5 (07)
10. a) List the advantages of using cache memory. CO5 (06)
 b) Consider a cache memory with 8 blocks of one word size and the following reference string of main memory blocks to be loaded into cache 18, 13, 128, 34, 46, 33, 247
 Demonstrate how the following cache mapping techniques will assign these blocks to cache:
 i) Direct mapping ii) Two-way set associative mapping iii) Fully associative mapping. CO5 (07)
 c) Discuss the hardware support required for interrupt technique. CO5 (07)
